

Week 1 Module – C & C++ Programming

Topics to be Covered

Students are required to Submit handwritten notes and practice for the following topics in **C and C++ Programming Language**:

1. **Syntax**
2. **Conditions and Loops**
3. **Programming Problems**
 - Solve the given **Star Pattern Questions**

1. Square

```
*****
*****
*****
*****
*****
```

2. Hollow Square

```
*****
*   *
*   *
*   *
*   *
*****
```

3. Triangle

```
*
**
***
****
*****
```

4. Inverted Triangle

```
*****
****
***
**
*
```

5. Pyramid

```
  *
 ***
*****
*****
*****
```

6. Inverted Pyramid

```
*****
*****
*****
***
*
```

7. Half Diamond

```
*
**
***
****
*****
****
***
**
*
```

8. Diamond

```
  *
 ***
*****
*****
*****
*****
***
  *
```

1. Square

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
    int i, j;
```

```
    clrscr();
```

```
    for (i = 1; i <= 5; i++)
```

```
    {
```

```
        for (j = 1; j <= 5; j++)
```

```
            printf ("*");
```

```
        printf ("\n");
```

```
    }
```

```
    getch();
```

```
}
```

```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    int i,j;
    clrscr();

    for(i=1;i<=5;i++)
    {
        for(j=1;j<=5;j++)
            printf("*");
        printf("\n");
    }
    getch();
}
```

16:2

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2. Hollow Square

```
#include <stdio.h>
#include <conio.h>
```

```
void main()
```

```
{
```

```
    int i, j;
```

```
    clrscr();
```

```
    for (i = 1; i <= 5; i++)
```

```
    {
```

```
        for (j = 1; j <= 5; j++)
```

```
        {
```

```
            if (i == 1 || i == 5 || j == 1 || j == 5)
```

```
                printf("*");
```

```
            else
```

```
                printf(" ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    getch();
```

```
}
```

```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    int i,j;
    clrscr();
    for(i=1;i<=5;i++)
    {
        for(j=1;j<=5;j++)
        {
            if(i==1||i==5||j==1||j==5)
                printf("k");
            else
                printf(" ");
        }
        printf("\n");
    }
    getch();
}
```

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3

Triangle

```
#include <stdio.h>
#include <conio.h>
```

```
void main()
```

```
{
```

```
    int i, j;
```

```
    clrscr();
```

```
    for (i = 1; i <= 5; i++)
```

```
    {
```

```
        for (j = 1; j <= i; j++)
```

```
            printf ("*");
```

```
        printf ("\n");
```

```
    }
```

```
    getch();
```

```
}
```



```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    int i,j,k;
    clrscr();

    for(i=1;i<=5;i++)
    {
        for(j=1;j<=i;j++)
            printf("%d",j);
        printf("\n");
    }
    getch();
}
```

*

—

4

Inverted Triangle

#include <stdio.h>

#include <conio.h>

void main()

{

int i, j;

clrscr();

for (i = 5; i >= 1; i--)

{

for (j = 1; j <= i; j++)

printf(" * ");

printf("\n");

}

getch();

}


```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    int i,j;
    clrscr();

    for(i=5; i>=1; i--)
    {
        for(j=1; j<=i; j++)
            printf("*");
        printf("\n");
    }
    getch();
}
```

—

5. pyramid

```
#include <stdio.h>
#include <conio.h>
```

```
void main()
```

```
{
```

```
int i, j, k;
```

```
clrscr();
```

```
for (i = 1; i <= 5; i++)
```

```
{
```

```
for (j = 1; j <= 5 - i; j++)
    printf(" ");
```

```
for (k = 1; k <= 2 * i - 1; k++)
    printf(" * ");
```

```
printf("\n");
```

```
}
```

```
getch();
```

```
}
```


[■] Save Search Run Compile Debug Project Opti
#include<stdio.h>
#include<conio.h>

```
void main()  
{  
    int i,j,k;  
    clrscr();  
  
    for(i=1;i<=5;i++)  
    {  
        for(j=1;j<=5-i;j++)  
            printf(" ");  
  
        for(k=1;k<=2*i-1;k++)  
            printf("%*");  
        printf("\n");  
    }  
    getch();  
}
```

*

—

6. Inverted Pyramid

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main ()
```

```
{
```

```
    int i, j, k;
```

```
    clrscr();
```

```
    for (i = 5; i >= 1; i--)
```

```
    {
```

```
        for (j = 1; j <= 5 - i; j++)
```

```
            printf(" ");
```

```
        for (k = 1; k <= 2 * i - 1; k++)
```

```
            printf("*");
```

```
        printf("\n");
```

```
    }
```

```
    getch();
```

```
}
```



```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    int i,j,k;
    clrscr();

    for(i=5;i>=1;i--)
    {
        for(j=1;j<=5-i;j++)
            printf(" ");

        for(k=1;k<=2*i-1;k++)
            printf("*");
        printf("\n");
    }
    getch();
}
```

19:66

XXXXXXXXXXXX

XXXXXXXXXX

XXXXXXX

XXXX

X

7. Half Diamond

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int i, j;
```

```
clrscr();
```

```
for (i = 1; i <= 5; i++)
```

```
{
```

```
for (j = 1; j <= i; j++)
```

```
printf("*");
```

```
printf("\n");
```

```
}
```

```
for (i = 4; i >= 1; i--)
```

```
{
```

```
for (j = 4; j <= i; j++)
```

```
printf("*");
```

```
printf("\n");
```

```
}
```

```
getch();
```

```
}
```



```
#include<stdio.h>
#include<conio.h>
```

```
void main()
```

```
{
```

```
    int i,j;
```

```
    clrscr();
```

```
    for(i=1;i<=5;i++)
```

```
    {
```

```
        for(j=1;j<=i;j++)
```

```
            printf("%c");
```

```
        printf("\n");
```

```
    }
```

```
    for(i=4;i>=1;i--)
```

```
    {
```

```
        for(j=1;j<=i;j++)
```

```
            printf("%c");
```

```
        printf("\n");
```

```
    }
```

```
    getch();
```

11:18

*

**

*

—

8.

Diamond

```
#include <stdio.h>
#include <conio.h>
void main()
```

```
{
    int i, j, k;
```

```
    clrscr();
```

```
    for (i = 1; i <= 5; i++)
```

```
    {
```

```
        for (j = 1; j <= 5 - i; j++)
```

```
            printf(" ");
```

```
        for (k = 1; k <= 2 * i - 1; k++)
```

```
            printf("*");
```

```
        printf("\n");
```

```
    }
```

```
    for (i = 4; i >= 1; i--)
```

```
    {
```

```
        for (j = 1; j <= 5 - i; j++)
```

```
            printf(" ");
```

```
        for (k = 1; k <= 2 * i - 1; k++)
```

```
            printf("*");
```

```
        printf("\n");
```




3

getch () ;

3

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```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    int i,j,k;
    clrscr();

    for(i=1;i<=5;i++)
    {
        for(j=1;j<=5-i;j++)
            printf(" ");
        for(k=1;k<=2*i-1;k++)
            printf("%s");

        printf("\n");
    }
    for(i=4;i>=1;i--)
    {
        for(j=1;j<=5-i;j++)
            printf(" ");
```

[■]

printf(" ");

for(k=1;k<=2*i-1;k++)

printf("*");

printf("\n");

}

getch();

*

*

—